



CALICUT UNIVERSITY
FOUR-YEAR UNDERGRADUATE PROGRAMME
(CU-FYUGP)
BA ARABIC

Programme	BA Arabic Language and Literature						
Course Code	ARA8CJ407 / ARA8MN407						
Course Title	Application of AI in Arabic Learning and Career						
Type of Course	DSC - Major						
Semester	VIII						
Academic Level	400-499						
Course Details	Credit	Lecture per week	Tutorial per week	Practicum per week	Total Hours	Marks: 100	
	4	3	-	2	75	Internal	External
Pre-requisites	Nil						
Course Summary	This course teaches students how to leverage AI tools to enhance Arabic language learning, academic writing, and research. Participants will explore AI-driven text generation, translation, speech recognition, and ethical practices to boost their research productivity and proficiency in Modern Standard Arabic (MSA) and regional dialects.						

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Set up and use AI environments for Arabic processing and writing.	U, Ap	C, P	Assignment / Written Examination / Practical
CO2	Evaluate leading AI tools for Arabic learning and research.	U, An	C	Assignments / Written Examination
CO3	Analyze AI's handling of Arabic linguistics (e.g., morphology, diacritics, dialects).	An	C	Quiz / Assignment / Written Examination
CO4	Apply AI for Arabic text generation, translation, and academic writing.	Ap	P	Assignments / Written Examination / Practical Examination
CO5	Design multimedia Arabic learning materials using AI.	Ap, Cr	P	Assignments / Written Examination / Practical Examination
CO6	Use AI for data and sentiment analysis in Arabic studies.	Ap	P	Assignments / Written Examination / Practical Examination
CO7	Ethically use AI for data collection, writing support, and plagiarism detection.	Ap, E	P	Assignments / Written Examination / Practical Examination
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge (F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs	Marks
I	AI Foundations and Arabic-Enabled Tools		10	16
	01	AI & NLP Basics for Arabic: Generative AI basics, overcoming Arabic linguistic challenges (RTL script, morphology, diacritics), environment setup, and prompt engineering.	3	
	02	AI Writing Assistants: Proofreading, grammar correction, diacritization (tashkeel), and rephrasing using tools like Qalam.ai, Lisan.ai, Sahehly.com, and Alnnahwi.com.	4	
	03	Speech & Pronunciation: Speech-to-text and pronunciation feedback for MSA and dialects using tools like Kateb.ai, Talkpal.ai, and SpeakPal.ai.	3	
II	Advanced AI Applications in Arabic Writing		15	20
	04	Generative AI for Text: Academic drafting, paraphrasing, vocabulary expansion, and creative writing using ChatGPT, Google Gemini, ALLaM, AceGPT, and Jais.	4	
	05	Translation & Multilingual Support: Arabic-English and dialect translation, handling cultural nuances, and post-editing using Google Translate, DeepL, Tarjama, and AceGPT.	5	
	06	OCR & Advanced Features: Arabic OCR for printed/handwritten text, sentiment analysis, plagiarism detection, and workflow integration.	6	
III	AI in Arabic Learning and Interactive Tools		10	20
	07	Interactive Learning Platforms: Conversational practice, personalized tutoring, and grammar drills using custom chatbots, Talkpal.ai, SpeakPal.ai, and Duolingo.	3	
	08	Data Processing & Analysis: Analysing linguistic data (text frequency, sentiment) and visualizing progress using AI add-ons in spreadsheets (Excel/Google Sheets).	3	
	09	Ethics & Practicalities: Academic integrity, citing AI, identifying model bias, and understanding AI limitations regarding cultural and religious nuances.	4	
IV	Internet, AI Resources, and Emerging Trends		10	14
	10	Online Research Resources: Using AI for literature searches, summarization, and reference management via e-libraries, databases, and MOOCs (SWAYAM, Edraak, Rwaq).	4	
	11	Presentations & Content Creation: Generating Arabic slide content/scripts, integrating visuals, and designing interactive modules for teaching or learning.	3	
	12	Emerging Trends: Multimodal AI, advancements in Arabic NLP (dialect handling, ALLaM/Jais models), case studies, and ethical curriculum integration.	3	

		Total	45	70 External + 10 Internal
V	Practical		30	20
	1	Hands-on with AI writing and generation tools for Arabic	10	
	2	Designing AI chatbots or prompts for Arabic learning modules	10	
	3	AI-based analysis and translation exercises in Arabic	10	

Mapping of COs with PSOs and POs:

	PSO 1	PSO 2	PSO 3	PSO 4	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	1	-	3	-	2	1	3	-	3	3	2	-	2
CO 2	2	1	3	-	2	1	3	-	3	3	2	-	1
CO 3	2	1	3	-	2	1	3	-	3	3	2	-	1
CO 4	1	-	3	-	1	-	-	-	3	3	-	-	1
CO 5	1	-	3	-	1	-	-	-	3	3	-	-	1
CO 6	1	-	3	-	1	-	-	-	3	3	-	-	1
CO 7	3	-	3	1	2	1	3	-	2	3	2	1	3

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Components of Internal Evaluation

Sl. No	Assessment Tool	First 4 (Theory) Modules	Fifth Module (Open Ended)
1	Assignment / Mini Project / AI Prompt Design	2	-
2	Seminar / Group Discussion / Presentation	3	-
3	Mid Term Examination / Test Paper	5	-
4	Total	10	-

Practical Examination (Total 20 Marks)

Sl. No	Question Type	Marks
1	AI-assisted Arabic text generation and editing using tools like ChatGPT	5
2	Designing and testing an AI prompt for Arabic writing feedback	5
3	Preparation of an AI-driven Arabic learning presentation or module	5
4	AI-based translation and analysis of Arabic content for accuracy	5
	Total	20

Mapping of COs to Assessment Tools:

	Midterm Exam	Assignment / Quiz / Mini Project	Seminar / Presentation	Practical Examination	End Semester Examinations
CO 1	✓	✓	✓		✓
CO 2	✓	✓	✓		✓
CO 3	✓	✓	✓		✓
CO 4	✓	✓	✓		✓
CO 5				✓	✓
CO 6				✓	✓
CO 7				✓	✓

Prescribed Textbook

1	د. محمد صالح. تي، د. عبد الجليل. ت، تطبيق الذكاء الاصطناعي في تعلم اللغة العربية وكتابتها: الأسس النظرية والتطبيقات العملية، مكتبة الهدى، كاليكوت، 2026 (قيد الإعداد / متوقع النشر في 2026).
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Additional Learning Materials

1	Wei Cai, Using Artificial Intelligence in Language Education: Principles and Prompting Activities, Bloomsbury Academic, 2025 (or latest edition)
2	Louise Ohashi, Mary Hillis, & Robert Dykes (Eds.), Artificial Intelligence in Our Language Learning Classrooms, Candlin & Mynard, 2025.
3	Priten Shah, AI and the Future of Education: Teaching in the Age of Artificial Intelligence, Wiley, 2023 (updated editions available).
4	Fang Pan (Ed.), AI in Language Teaching, Learning, and Assessment, 2024.
5	Mozah H. Alkaabi & Asma Saeed Almaamari, "Generative AI Implementation and Assessment in Arabic Language Teaching" (article/chapter in International Journal of Online Pedagogy and Course Design), IGI Global, 2025 (relevant open-access insights on GenAI in Arabic classrooms).
6	Recent Advances in NLP: The Case of Arabic Language, Springer, 2020 (updated perspectives; foundational for Arabic NLP challenges in AI tools).
7	Various open resources/articles: e.g., "Artificial Intelligence in Arabic Language Education: Current Applications, Challenges, and Future Perspectives" (2025 studies/articles on ResearchGate or journals for Arabic-specific AI applications)
8	M. Abd Elaziz et al. (Eds.), Recent Advances in NLP: The Case of Arabic Language, Springer, 2020 (Studies in Computational Intelligence series).
9	Adnan Badran, Elias Baydoun, Sandra Hillman, & Joelle Mesmar (Eds.), Higher Education in the Arab World: Artificial Intelligence, Springer, 2026.